AP CALCULUS AB

Unit 1 • Lesson 1.7 practice

Lesson 1.7 • Selecting Procedures for Determining Limits

4 multiple-choice questions and 1 free-response question

Question 1

$\lim_{x\to 3} \dfrac{x+1}{(x-3)^2}$ is

(A) $+\infty$

(B) $-\infty$

(C) 0

(D) 4

Question 2

$\lim_{x\to 0} \dfrac{x^2+5x}{x}$ is

(A) 0

(B) 5

(C) 1

(D) It does not exist.

Question 3

$\lim_{x\to 1} \dfrac{x^2+1}{x+1}$ is

(A) 0

(B) 2

(C) 1

(D) It does not exist.

Question 4

$\lim_{x\to 4} \dfrac{|x-4|}{x-4}$ is

(A) 1

(B) -1

(C) 0

(D) It does not exist.

Question 5 (Free response)

For each limit, name the method you choose and evaluate the limit.

(a) $\lim_{x\to -1} \dfrac{x^2+2x+1}{x+1}$

(b) $\lim_{x\to 2^+} \dfrac{x}{2-x}$

(c) $\lim_{x\to 9} \dfrac{x-9}{\sqrt{x}-3}$

(d) $\lim_{x\to 0} \dfrac{\sin 2x}{\sin 3x}$